Assignment -IV

Engineering Graphics & Design (CE 101)

Projection of Planes

1. A rectangular plane ***ABCD*** is resting on the ground on its edge ***AB*** inclined at 30° to V.P and its surface inclined at 45° to H.P. Draw the projection of the plane if ***AB*** = ***CD*** = 30 mm and ***BC*** = ***DA*** = 50 mm
2. A circular plane of diameter 50 mm is resting on V.P on one end ***A*** of its diameter ***AB***. The surface of the plane is inclined at 45° to V.P and the diameter ***AB*** inclined at 30° to H.P. Draw the projections of the circle.
3. A regular pentagon of side 25 mm is resting on one of its edges on H.P inclined at 30° to V.P and the surface inclined at 45° to H.P. Draw its projections.
4. The top view of an isosceles triangle of base 50 mm and altitude 70 mm appears as an equilateral triangle of side 50 mm. The base of the triangle is inclined at 45° to V.P. Draw its projections
5. ***ABCD*** is a square of side 50 mm each. It is resting on H.P at the end ***A*** of its diagonal ***AC***. The surface of the plane is inclined at 45° to H.P. The diagonal *BD* is parallel to H.P and inclined at 30° to V.P. Draw the projections of the plane.
6. The top view of an elliptical plane inclined at 45° to H.P is a circle of diameter 50 mm. Find the true shape of the plane and also draw its front view if its major axis is inclined at 30° to V.P.
7. ***ABCD*** is a parallelogram with ***AB*** = ***CD*** = 30 mm while ***BC*** = ***DA*** = 50 mm. It is resting on the ground at the end A of its diagonal AC which is inclined at 30° to V.P and the surface inclined at 45° to H.P. Draw its projections.
8. A regular hexagonal plane of side 25 mm is resting on V.P on one its corner. The surface of the plane is inclined at 30° to V.P and the diagonal passing the resting corner on V.P inclined at 45° to H.P. Draw its projections.